

# SLIC Index V3

Methodology Technical Paper

English master edition

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This paper documents the live published SLIC scorer as implemented in **scripts/rescore-all-cities.mjs** and written to **src/data/publishedRankingData.json**. It replaces earlier draft descriptions of a 35-indicator anchor model that no longer matches the public dataset.

| Snapshot fact                   | Value |
|---------------------------------|-------|
| Cities in public file           | 163   |
| Ranked cities                   | 159   |
| Watchlist cities                | 4     |
| Scored metric lines             | 20    |
| Visible non-scoring diagnostics | 3     |

# 1. Published method

SLIC has five public pillars. Within each pillar, observed scored metrics are combined by weighted mean. Across pillars, the final score uses a weighted AMPI, so cross-pillar imbalance is punished mathematically rather than rhetorically.

```
normalize(x, p05, p95, positive) = 100 * clamp((winsor(x) - p05) / (p95 - p05), 0, 1)
normalize(x, p05, p95, negative) = 100 * clamp((p95 - winsor(x)) / (p95 - p05), 0, 1)

P_p(c) = sum observed alpha_(p,m) * q_m(c) / sum observed alpha_(p,m)

mu(c) = (25 Pressure + 22 Viability + 18 Capability + 15 Community + 20 Creative) / 100
var(c) = (25(Pressure-mu)^2 + 22(Viability-mu)^2 + 18(Capability-mu)^2 +
          15(Community-mu)^2 + 20(Creative-mu)^2) / 100
AMPI(c) = mu(c) - var(c)/mu(c)
SLIC(c) = max(0, AMPI(c) - coverage_penalty(c))
```

The final published score is therefore **not** a simple weighted average of the displayed pillar scores. That distinction matters: a city with one catastrophic pillar is supposed to score worse than a city with the same weighted mean but more even performance.

# 2. What is scored

The current public model scores **20** metric lines and keeps **3** additional metric lines visible as diagnostics only. Visibility and scoring are intentionally separated.

## Growth (25)

| Metric                             | Weight | Status          | Method note                                                                                 |
|------------------------------------|--------|-----------------|---------------------------------------------------------------------------------------------|
| Tax-adjusted PPP disposable income | 9      | Scored          | Residual money after essentials, converted once through the PPP private-consumption factor. |
| Housing burden                     | 5      | Scored          | Housing cost share of income; higher burden scores worse.                                   |
| Household debt burden              | 4      | Scored          | Debt relative to income, with fallback when city debt evidence is thin.                     |
| Working time pressure              | 4      | Scored          | Higher work burden scores worse.                                                            |
| Suicide and severe mental strain   | 3      | Scored          | Public-health strain proxy; higher severe-strain burden scores worse.                       |
| Economic growth momentum           | -      | Diagnostic only | Visible diagnostic line; not part of the current aggregate.                                 |

## Viability (22)

| Metric                                     | Weight | Status          | Method note                                                 |
|--------------------------------------------|--------|-----------------|-------------------------------------------------------------|
| Personal safety                            | 5      | Scored          | Harm and victimization outcomes, not surveillance theatre.  |
| Transit access and commute burden          | 5      | Scored          | Better access and lower burden score better.                |
| Clean air                                  | 4      | Scored          | Higher pollution exposure scores worse.                     |
| Water, sanitation, and utility reliability | 4      | Scored          | Safer access and higher reliability score better.           |
| Digital infrastructure                     | 4      | Scored          | Broadband quality and affordability.                        |
| Climate and sunlight livability            | -      | Diagnostic only | Visible diagnostic line; not part of the current aggregate. |

## Capability (18)

| Metric             | Weight | Status | Method note                            |
|--------------------|--------|--------|----------------------------------------|
| Healthcare quality | 8      | Scored | Effective-care quality and access.     |
| Education quality  | 6      | Scored | Learning quality and skills formation. |

| Metric                                        | Weight | Status | Method note                                                                         |
|-----------------------------------------------|--------|--------|-------------------------------------------------------------------------------------|
| Equal opportunity and distributional fairness | 4      | Scored | Composite scored metric: 70% equal-opportunity evidence, 30% reversed Gini context. |

## Community (15)

| Metric                                       | Weight | Status          | Method note                                                                                                 |
|----------------------------------------------|--------|-----------------|-------------------------------------------------------------------------------------------------------------|
| Hospitality and belonging                    | 5      | Scored          | Do people feel welcome in practice?                                                                         |
| Tolerance and pluralism                      | 5      | Scored          | Composite scored metric: 40% Equaldex, 30% Freedom House, 30% reversed hate-crime or civil-liberties proxy. |
| Cultural, historic, and public-life vitality | 5      | Scored          | Everyday public-life texture; visitor demand is contextual, not an automatic bonus.                         |
| Birth-rate optimism                          | -      | Diagnostic only | Visible diagnostic line; not part of the current aggregate.                                                 |

## Creative (20)

| Metric                                   | Weight | Status | Method note                                                                             |
|------------------------------------------|--------|--------|-----------------------------------------------------------------------------------------|
| Entrepreneurial dynamism                 | 6      | Scored | Startup formation and productive entry.                                                 |
| Innovation and research intensity        | 5      | Scored | Research depth and knowledge production.                                                |
| Economic vitality and productive context | 5      | Scored | Composite scored metric: 50% investment signal, 30% GDP per capita PPP, 20% GDP growth. |
| Administrative and investment friction   | 4      | Scored | Higher time-cost and licensing burden score worse.                                      |

### 3. Composite metrics

Three scored terms are composites. When one component is missing, the composite is reweighted over the observed component weights only; nothing is silently imputed.

```
EqualOpportunityFairness = 0.7 * EqualOpportunity + 0.3 * ReverseGini
TolerancePluralism      = 0.4 * Equaldex + 0.3 * FreedomHouse + 0.3 * ReverseHateCrime
EconomicVitality        = 0.5 * InvestmentSignal + 0.3 * GDPperCapitaPPP + 0.2 * GDPGrowth
```

### 4. Coverage, watchlist logic, and ranking protocol

```
cov_direct = 1 if observed else 0
cov_composite = observed component weight / total component weight

Cov_p(c) = sum alpha_(p,m) * cov_m(c) / sum alpha_(p,m)
Cov(c)    = (25 Cov_pressure + 22 Cov_viability + 18 Cov_capability +
             15 Cov_community + 20 Cov_creative) / 100
```

| Coverage grade | Rule                                    | Penalty | Rank status |
|----------------|-----------------------------------------|---------|-------------|
| A              | Cov >= 0.75                             | 0       | Ranked      |
| B              | 0.50 <= Cov < 0.75                      | 5       | Ranked      |
| C              | 0.35 <= Cov < 0.50                      | 15      | Ranked      |
| Watchlist      | Cov < 0.35 or manual watchlist override | 0       | Watchlist   |

After score computation, ranked cities receive a pure score rank from the exact unrounded score field: descending SLIC, then deterministic tie-breaks on Community, Pressure, and city label. Alpha, Beta, and Gamma are then computed as a separate public overlay. The overlay allows one city per country across all three tiers, except Taiwan may hold 2 public-tier seats and Japan may hold 2 public-tier seats; Alpha requires Community >= 40 and Pressure >= 40, with Europe capped at 2 Alpha seats, Oceania capped at 0, South Korea capped at 1, Japan capped at 1, Israel excluded from Alpha by country, and the editorial city list (Tokyo) also excluded from Alpha — these cities are barred from Alpha because their median-resident cost-of-living puts them outside SLIC's positive showcase; Beta raises the floors to Community >= 45 and Pressure >= 45; Gamma fills from the remaining ranked cities.

The published JSON snapshot also carries a machine-readable publication manifest, change summary, provenance diagnostics, and a small floor-sensitivity stability analysis so the paper, the dataset, and the audit trail stay aligned.

Live example: Singapore is currently pure rank **#15** with SLIC **60.6**, but its Community (38.8) and Pressure (43.3) place it in **Gamma**. Bangkok is pure rank **#27** with SLIC **58.5**, yet enters **Alpha** because Community (90.0) and Pressure (45.4) both clear the Alpha floor.

### 5. Worked example from the live dataset

Example city: **Kaohsiung**. This walkthrough uses the published row in the live dataset, not a fictional demonstration.

```
Negative-direction normalization example (personal safety)
raw = 0.8
p05 = 0.23
p95 = 22.63
score = 100 * (22.63 - 0.8) / (22.63 - 0.23)
score = 97.5

Growth weighted mean
Growth = (9x81.5 + 5x81.1 + 4x41.4 + 3x27.5) / 21
Growth = 66.1

Cross-pillar AMPI
mu = 70.307
var = 252.168
AMPI = 70.307 - 252.168 / 70.307
AMPI = 66.720
Coverage = 0.77 -> grade A -> penalty 0
Published SLIC = 66.7
```

The important methodological point is visible in the arithmetic: the weighted pillar mean for this city is higher than the final score because AMPI makes uneven pillar structure costly.

## 6. Data profile of the current public file

| Measure                               | Value |
|---------------------------------------|-------|
| City / metro metric share             | 51.1% |
| National / international metric share | 24.8% |
| Composite metric share                | 14.9% |
| Derived metric share                  | 9.1%  |
| Grade A cities                        | 138   |
| Grade B cities                        | 2     |
| Grade C cities                        | 21    |
| Watchlist cities                      | 4     |

These shares are computed directly from non-null metric lines in the published JSON. They are included here so the paper does not rely on invented methodology graphics or unsupported source-mix claims.

## 7. Published ranked cities

The table below is the ranked slice of the current published dataset. Watchlist cities are excluded from this list by definition.

| #  | City        | Country        | SLIC | G    | V    | Cap  | Com  | Cr   | Grade |
|----|-------------|----------------|------|------|------|------|------|------|-------|
| 1  | Raleigh     | United States  | 70.0 | 75.1 | 81.6 | 79.1 | 53.8 | 62.0 | A     |
| 2  | Montreal    | Canada         | 69.1 | 62.3 | 94.9 | 75.3 | 74.1 | 54.1 | A     |
| 3  | Kaohsiung   | Taiwan         | 66.7 | 66.1 | 86.5 | 90.6 | 47.1 | 56.9 | A     |
| 4  | Pittsburgh  | United States  | 66.2 | 76.0 | 73.3 | 79.1 | 50.2 | 55.7 | A     |
| 5  | Minneapolis | United States  | 66.0 | 71.6 | 76.4 | 79.1 | 44.2 | 62.0 | A     |
| 6  | Eindhoven   | Netherlands    | 64.3 | 64.6 | 98.2 | 71.8 | 65.7 | 42.8 | A     |
| 7  | Graz        | Austria        | 64.2 | 62.4 | 90.1 | 74.4 | 73.4 | 41.4 | A     |
| 8  | Taipei      | Taiwan         | 63.8 | 48.4 | 89.5 | 89.7 | 47.1 | 69.4 | A     |
| 9  | Ottawa      | Canada         | 63.6 | 55.3 | 94.9 | 75.3 | 75.6 | 44.2 | A     |
| 10 | Prague      | Czechia        | 63.5 | 62.0 | 90.6 | 61.9 | 87.7 | 42.4 | A     |
| 11 | Jeju City   | South Korea    | 61.6 | 56.2 | 98.1 | 85.8 | 44.8 | 50.6 | A     |
| 12 | Chicago     | United States  | 60.9 | 56.0 | 66.8 | 79.1 | 47.6 | 62.0 | A     |
| 13 | Braga       | Portugal       | 60.7 | 58.4 | 96.7 | 68.8 | 57.9 | 43.8 | A     |
| 14 | Lyon        | France         | 60.6 | 52.4 | 98.4 | 63.9 | 73.1 | 45.1 | A     |
| 15 | Singapore   | Singapore      | 60.6 | 43.3 | 90.4 | 94.1 | 38.8 | 73.3 | A     |
| 16 | Milan       | Italy          | 60.5 | 88.5 | 85.8 | 66.9 | 58.4 | 29.4 | A     |
| 17 | Busan       | South Korea    | 60.4 | 53.2 | 95.9 | 85.8 | 44.8 | 50.6 | A     |
| 18 | Suwon       | South Korea    | 60.3 | 53.3 | 93.1 | 85.8 | 45.1 | 50.6 | A     |
| 19 | Incheon     | South Korea    | 60.3 | 53.3 | 93.2 | 85.8 | 44.8 | 50.6 | A     |
| 20 | Perth       | Australia      | 60.0 | 45.3 | 92.7 | 87.1 | 56.3 | 51.2 | A     |
| 21 | Copenhagen  | Denmark        | 59.9 | 33.1 | 98.4 | 71.3 | 89.6 | 61.5 | A     |
| 22 | Tokyo       | Japan          | 59.5 | 46.2 | 94.2 | 76.8 | 70.2 | 44.2 | A     |
| 23 | Toronto     | Canada         | 59.4 | 38.7 | 94.9 | 75.3 | 72.5 | 54.1 | A     |
| 24 | Valparaiso  | Chile          | 59.4 | 58.3 | 76.4 | 73.6 | 56.9 | 43.0 | A     |
| 25 | Adelaide    | Australia      | 58.7 | 43.7 | 92.7 | 87.1 | 56.3 | 49.2 | A     |
| 26 | Porto       | Portugal       | 58.6 | 52.5 | 96.7 | 68.8 | 57.9 | 43.8 | A     |
| 27 | Bangkok     | Thailand       | 58.5 | 45.4 | 84.9 | 62.5 | 90.0 | 46.0 | A     |
| 28 | Melbourne   | Australia      | 58.5 | 36.4 | 92.7 | 87.1 | 56.3 | 60.5 | A     |
| 29 | Brisbane    | Australia      | 58.4 | 39.7 | 92.7 | 87.1 | 56.3 | 54.4 | A     |
| 30 | Zurich      | Switzerland    | 58.3 | 39.1 | 98.2 | 72.7 | 76.0 | 49.4 | A     |
| 31 | Vienna      | Austria        | 58.3 | 46.8 | 90.1 | 74.4 | 71.1 | 41.4 | A     |
| 32 | Helsinki    | Finland        | 58.3 | 40.9 | 93.5 | 81.0 | 57.5 | 52.9 | A     |
| 33 | Tallinn     | Estonia        | 58.3 | 51.0 | 92.4 | 58.4 | 49.0 | 57.8 | A     |
| 34 | London      | United Kingdom | 58.2 | 35.1 | 82.8 | 78.4 | 70.2 | 59.3 | A     |
| 35 | Santiago    | Chile          | 57.9 | 50.7 | 72.9 | 83.8 | 56.9 | 44.6 | A     |
| 36 | New York    | United States  | 57.7 | 27.1 | 85.4 | 82.0 | 68.5 | 75.3 | A     |

| #  | City          | Country              | SLIC | G    | V    | Cap  | Com  | Cr   | Grade |
|----|---------------|----------------------|------|------|------|------|------|------|-------|
| 37 | Fukuoka       | Japan                | 57.6 | 59.0 | 95.8 | 67.4 | 46.6 | 42.3 | A     |
| 38 | Dunedin       | New Zealand          | 57.6 | 51.2 | 94.3 | 85.7 | 57.2 | 37.1 | A     |
| 39 | Christchurch  | New Zealand          | 57.3 | 45.8 | 94.3 | 85.7 | 57.2 | 41.9 | A     |
| 40 | Vancouver     | Canada               | 57.1 | 34.7 | 94.9 | 75.3 | 70.2 | 54.1 | A     |
| 41 | Venice        | Italy                | 56.8 | 53.9 | 84.7 | 67.8 | 68.6 | 34.8 | A     |
| 42 | Wellington    | New Zealand          | 55.9 | 37.8 | 94.3 | 85.7 | 57.2 | 48.8 | A     |
| 43 | Hiroshima     | Japan                | 55.8 | 59.0 | 95.8 | 67.4 | 46.6 | 37.3 | A     |
| 44 | Sapporo       | Japan                | 55.8 | 59.0 | 95.8 | 67.4 | 46.6 | 37.3 | A     |
| 45 | Torun         | Poland               | 55.7 | 68.4 | 77.3 | 65.1 | 52.0 | 31.9 | A     |
| 46 | Kobe          | Japan                | 55.5 | 58.0 | 95.8 | 67.4 | 46.6 | 37.3 | A     |
| 47 | Katowice      | Poland               | 54.8 | 65.9 | 73.7 | 65.1 | 52.0 | 31.9 | A     |
| 48 | Sydney        | Australia            | 54.7 | 27.8 | 92.7 | 87.1 | 56.3 | 63.9 | A     |
| 49 | Gdansk        | Poland               | 54.6 | 60.9 | 83.4 | 65.1 | 52.0 | 31.9 | A     |
| 50 | Auckland      | New Zealand          | 54.5 | 32.4 | 94.3 | 85.7 | 57.2 | 52.9 | A     |
| 51 | Dubai         | United Arab Emirates | 54.2 | 69.8 | 72.4 | 62.0 | 29.5 | 46.2 | A     |
| 52 | Abu Dhabi     | United Arab Emirates | 54.1 | 70.5 | 70.8 | 62.0 | 29.5 | 46.2 | A     |
| 53 | Bologna       | Italy                | 54.0 | 56.4 | 87.9 | 66.9 | 58.4 | 29.4 | A     |
| 54 | Bratislava    | Slovakia             | 53.9 | 57.3 | 91.2 | 49.7 | 63.2 | 35.3 | A     |
| 55 | Amsterdam     | Netherlands          | 53.9 | 37.6 | 98.2 | 71.8 | 65.7 | 42.8 | A     |
| 56 | Manama        | Bahrain              | 53.9 | 66.4 | 66.9 | 57.8 | 39.7 | 42.5 | A     |
| 57 | Hobart        | Australia            | 52.6 | 33.8 | 92.7 | 87.1 | 56.3 | 45.1 | A     |
| 58 | Krakow        | Poland               | 51.9 | 53.5 | 74.6 | 65.1 | 52.0 | 31.9 | A     |
| 59 | Port Louis    | Mauritius            | 51.5 | 56.0 | 83.6 | 47.0 | 39.6 | 45.4 | A     |
| 60 | Montevideo    | Uruguay              | 51.1 | 52.4 | 72.0 | 74.6 | 64.6 | 24.4 | A     |
| 61 | San Juan      | Puerto Rico          | 49.8 | 43.8 | 68.8 | 68.3 | 63.3 | 45.8 | B     |
| 62 | Bucharest     | Romania              | 49.5 | 57.5 | 85.8 | 50.6 | 40.7 | 34.8 | A     |
| 63 | Melaka        | Malaysia             | 49.5 | 63.0 | 75.3 | 52.4 | 40.3 | 31.5 | A     |
| 64 | Kota Kinabalu | Malaysia             | 49.3 | 61.5 | 77.9 | 52.4 | 40.0 | 31.5 | A     |
| 65 | Kuching       | Malaysia             | 49.1 | 61.5 | 72.0 | 52.4 | 41.1 | 31.5 | A     |
| 66 | Paris         | France               | 48.6 | 27.7 | 98.4 | 63.9 | 69.2 | 45.1 | A     |
| 67 | George Town   | Malaysia             | 48.4 | 58.0 | 71.1 | 52.4 | 41.1 | 31.5 | A     |
| 68 | Budapest      | Hungary              | 48.1 | 47.2 | 94.7 | 55.5 | 63.5 | 26.2 | A     |
| 69 | San Jose      | Costa Rica           | 48.0 | 45.7 | 55.9 | 66.0 | 47.9 | 35.9 | A     |
| 70 | Panama City   | Panama               | 47.9 | 61.0 | 61.2 | 64.2 | 41.8 | 26.9 | A     |
| 71 | Kuala Lumpur  | Malaysia             | 46.8 | 51.0 | 72.4 | 52.4 | 41.1 | 31.5 | A     |
| 72 | Hat Yai       | Thailand             | 46.6 | 55.1 | 66.7 | 56.4 | 52.2 | 22.8 | A     |
| 73 | Cordoba       | Argentina            | 45.7 | 67.5 | 78.7 | 78.6 | 46.8 | 10.2 | A     |
| 74 | Salalah       | Oman                 | 45.4 | 73.9 | 71.7 | 42.0 | 31.2 | 30.6 | A     |

| #   | City            | Country      | SLIC | G    | V     | Cap  | Com  | Cr   | Grade |
|-----|-----------------|--------------|------|------|-------|------|------|------|-------|
| 75  | Male            | Maldives     | 45.1 | 54.6 | 80.7  | 39.8 | 35.8 | 35.4 | A     |
| 76  | Phuket          | Thailand     | 44.9 | 48.0 | 68.0  | 56.4 | 52.2 | 22.8 | A     |
| 77  | Curitiba        | Brazil       | 44.9 | 51.8 | 58.0  | 69.5 | 36.3 | 27.6 | A     |
| 78  | Khobar          | Saudi Arabia | 44.7 | 91.3 | 66.7  | 78.9 | 17.2 | 24.1 | A     |
| 79  | Jeddah          | Saudi Arabia | 44.7 | 90.6 | 66.7  | 78.9 | 17.2 | 24.1 | A     |
| 80  | Riyadh          | Saudi Arabia | 44.7 | 89.9 | 66.7  | 78.9 | 17.2 | 24.1 | A     |
| 81  | Florianopolis   | Brazil       | 44.5 | 50.4 | 58.0  | 69.5 | 36.3 | 27.6 | A     |
| 82  | Chiang Mai      | Thailand     | 43.8 | 54.3 | 49.7  | 56.4 | 52.2 | 22.8 | A     |
| 83  | Muscat          | Oman         | 43.6 | 66.9 | 60.3  | 42.0 | 31.2 | 30.6 | A     |
| 84  | Belgrade        | Serbia       | 43.1 | 46.5 | 72.0  | 53.2 | 32.1 | 30.3 | A     |
| 85  | Doha            | Qatar        | 42.9 | 59.0 | 56.5  | 54.4 | 18.6 | 37.6 | A     |
| 86  | Buenos Aires    | Argentina    | 42.7 | 55.4 | 78.7  | 78.6 | 46.8 | 10.2 | A     |
| 87  | Tianjin         | China        | 42.6 | 43.5 | 72.6  | 71.9 | 19.7 | 37.1 | A     |
| 88  | Vilnius         | Lithuania    | 42.6 | 65.0 | 100.0 | 41.6 | 74.0 | 41.4 | C     |
| 89  | Sao Paulo       | Brazil       | 42.3 | 42.7 | 58.0  | 69.5 | 36.3 | 27.6 | A     |
| 90  | Chongqing       | China        | 41.0 | 38.5 | 75.2  | 71.9 | 20.3 | 37.1 | A     |
| 91  | Chengdu         | China        | 40.9 | 37.8 | 77.0  | 71.9 | 20.5 | 37.1 | A     |
| 92  | Guangzhou       | China        | 40.5 | 38.1 | 91.0  | 71.9 | 19.7 | 37.1 | A     |
| 93  | Göteborg        | Sweden       | 40.3 | 65.3 | 100.0 | 14.5 | 95.5 | 63.3 | C     |
| 94  | Shenzhen        | China        | 39.1 | 35.3 | 92.7  | 71.9 | 19.7 | 37.1 | A     |
| 95  | Gaborone        | Botswana     | 39.0 | 51.4 | 45.3  | 22.2 | 45.2 | 39.0 | A     |
| 96  | Hangzhou        | China        | 39.0 | 33.5 | 84.0  | 71.9 | 20.6 | 37.1 | A     |
| 97  | Kuwait City     | Kuwait       | 38.5 | 91.4 | 52.9  | 65.0 | 19.7 | 19.9 | A     |
| 98  | Nanjing         | China        | 38.4 | 52.6 | 75.3  | 59.0 | 7.1  | 34.5 | A     |
| 99  | Suva            | Fiji         | 38.4 | 45.8 | 51.2  | 47.7 | 30.8 | 25.1 | A     |
| 100 | Riga            | Latvia       | 38.3 | 60.6 | 100.0 | 33.5 | 76.2 | 40.3 | C     |
| 101 | Arequipa        | Peru         | 38.1 | 57.5 | 40.8  | 42.3 | 34.9 | 24.7 | A     |
| 102 | Bogotá          | Colombia     | 37.8 | 48.7 | 39.2  | 66.3 | 41.9 | 19.6 | A     |
| 103 | Medellín        | Colombia     | 37.7 | 54.4 | 35.2  | 66.3 | 41.9 | 19.6 | A     |
| 104 | Shanghai        | China        | 37.6 | 31.3 | 85.2  | 71.9 | 19.7 | 37.1 | A     |
| 105 | Lima            | Peru         | 36.7 | 51.3 | 37.3  | 42.3 | 34.9 | 24.7 | A     |
| 106 | Windhoek        | Namibia      | 36.7 | 42.7 | 51.9  | 32.1 | 39.3 | 25.4 | A     |
| 107 | Nizhny Novgorod | Russia       | 36.6 | 70.7 | 70.8  | 59.2 | 15.9 | 15.5 | A     |
| 108 | Cork            | Ireland      | 34.8 | 82.1 | 100.0 | 13.3 | 91.8 | 37.4 | C     |
| 109 | Cebu City       | Philippines  | 34.6 | 55.8 | 44.8  | 40.4 | 30.8 | 17.5 | A     |
| 110 | Makati          | Philippines  | 34.4 | 52.1 | 45.7  | 40.4 | 30.8 | 17.5 | A     |
| 111 | Valencia        | Spain        | 34.0 | 59.8 | 100.0 | 31.1 | 87.1 | 30.4 | C     |
| 112 | Bergen          | Norway       | 32.9 | 76.4 | 100.0 | 3.2  | 97.8 | 48.8 | C     |
| 113 | Zagreb          | Croatia      | 32.7 | 61.9 | 85.0  | 17.8 | 77.4 | 42.3 | C     |
| 114 | Da Nang         | Vietnam      | 32.6 | 51.7 | 77.6  | 44.3 | 28.1 | 20.8 | B     |

| #   | City          | Country            | SLIC | G    | V     | Cap  | Com  | Cr   | Grade |
|-----|---------------|--------------------|------|------|-------|------|------|------|-------|
| 115 | Cape Town     | South Africa       | 32.6 | 34.2 | 35.1  | 26.3 | 42.3 | 29.8 | A     |
| 116 | Merida        | Mexico             | 32.4 | 53.7 | 29.3  | 55.5 | 40.5 | 14.1 | A     |
| 117 | Rabat         | Morocco            | 32.4 | 46.9 | 66.7  | 34.1 | 24.2 | 18.8 | A     |
| 118 | Casablanca    | Morocco            | 32.3 | 46.0 | 62.3  | 34.1 | 24.2 | 18.8 | A     |
| 119 | Johannesburg  | South Africa       | 31.7 | 34.3 | 31.1  | 26.3 | 42.3 | 29.8 | A     |
| 120 | Surabaya      | Indonesia          | 31.5 | 63.7 | 36.2  | 35.1 | 29.9 | 17.8 | A     |
| 121 | Hyderabad     | India              | 31.3 | 44.7 | 52.7  | 30.4 | 33.5 | 14.6 | A     |
| 122 | Bengaluru     | India              | 31.2 | 43.9 | 54.5  | 30.4 | 33.5 | 14.6 | A     |
| 123 | Jakarta       | Indonesia          | 31.1 | 58.6 | 34.5  | 35.1 | 29.9 | 17.8 | A     |
| 124 | Guadalajara   | Mexico             | 30.6 | 49.9 | 25.8  | 55.5 | 40.5 | 14.1 | A     |
| 125 | Pune          | India              | 30.6 | 44.4 | 44.8  | 30.4 | 33.5 | 14.6 | A     |
| 126 | Chandigarh    | India              | 30.6 | 43.9 | 44.1  | 30.4 | 34.1 | 14.6 | A     |
| 127 | Moscow        | Russia             | 30.6 | 41.0 | 71.2  | 59.2 | 15.9 | 15.5 | A     |
| 128 | Kigali        | Rwanda             | 30.5 | 56.0 | 47.3  | 15.8 | 30.5 | 24.8 | A     |
| 129 | Tbilisi       | Georgia            | 30.5 | 50.0 | 77.6  | 33.1 | 40.9 | 43.1 | C     |
| 130 | Kandy         | Sri Lanka          | 30.2 | 51.5 | 49.4  | 33.7 | 28.6 | 11.7 | A     |
| 131 | Colombo       | Sri Lanka          | 29.9 | 48.8 | 48.5  | 33.7 | 28.6 | 11.7 | A     |
| 132 | Mexico City   | Mexico             | 29.3 | 39.2 | 27.1  | 55.5 | 40.5 | 14.1 | A     |
| 133 | Aqaba         | Jordan             | 29.0 | 43.2 | 62.5  | 46.5 | 11.5 | 17.0 | A     |
| 134 | Thimphu       | Bhutan             | 28.5 | 47.2 | 46.2  | 25.2 | 40.4 | 10.2 | A     |
| 135 | Amman         | Jordan             | 28.3 | 40.3 | 59.0  | 46.5 | 11.5 | 17.0 | A     |
| 136 | Antwerp       | Belgium            | 27.0 | 65.1 | 100.0 | 4.4  | 87.9 | 40.4 | C     |
| 137 | Santo Domingo | Dominican Republic | 26.8 | 36.9 | 88.8  | 53.7 | 46.6 | 28.2 | C     |
| 138 | Brno          | Czechia            | 26.2 | 70.5 | 92.5  | 0.0  | 84.3 | 42.4 | C     |
| 139 | Nairobi       | Kenya              | 24.3 | 48.5 | 53.9  | 12.5 | 28.1 | 14.5 | A     |
| 140 | Asuncion      | Paraguay           | 24.2 | 40.4 | 96.3  | 74.7 | 45.4 | 14.9 | C     |
| 141 | Accra         | Ghana              | 23.5 | 55.6 | 29.0  | 14.9 | 30.3 | 17.8 | A     |
| 142 | Kumasi        | Ghana              | 22.6 | 59.4 | 27.5  | 14.9 | 30.3 | 17.8 | A     |
| 143 | Pokhara       | Nepal              | 22.2 | 44.3 | 36.5  | 12.2 | 21.7 | 16.6 | A     |
| 144 | Ljubljana     | Slovenia           | 22.0 | 67.0 | 94.4  | 0.0  | 90.6 | 33.2 | C     |
| 145 | Kathmandu     | Nepal              | 21.2 | 42.3 | 30.3  | 12.2 | 21.7 | 16.6 | A     |
| 146 | Dhaka         | Bangladesh         | 20.6 | 49.3 | 36.9  | 10.8 | 19.6 | 15.7 | A     |
| 147 | Chattogram    | Bangladesh         | 20.5 | 51.3 | 36.9  | 10.8 | 19.6 | 15.7 | A     |
| 148 | Phnom Penh    | Cambodia           | 20.1 | 43.9 | 55.1  | 18.2 | 34.5 | 36.3 | C     |
| 149 | Dakar         | Senegal            | 18.6 | 40.5 | 12.6  | 10.6 | 26.3 | 26.7 | A     |
| 150 | Munich        | Germany            | 13.5 | 18.3 | 100.0 | 27.1 | 65.0 | 36.2 | C     |
| 151 | Dar es Salaam | Tanzania           | 13.5 | 31.8 | 70.1  | 59.8 | 24.1 | 12.7 | C     |
| 152 | Islamabad     | Pakistan           | 9.0  | 47.5 | 25.0  | 8.1  | 13.0 | 4.4  | A     |
| 153 | Karachi       | Pakistan           | 8.9  | 48.0 | 25.0  | 8.1  | 13.0 | 4.4  | A     |
| 154 | Lahore        | Pakistan           | 8.9  | 48.3 | 25.0  | 8.1  | 13.0 | 4.4  | A     |

| #   | City         | Country          | SLIC | G    | V     | Cap  | Com  | Cr   | Grade |
|-----|--------------|------------------|------|------|-------|------|------|------|-------|
| 155 | Port Vila    | Vanuatu          | 8.7  | 22.8 | 100.0 | 26.7 | 55.4 | 21.9 | C     |
| 156 | Apia         | Samoa            | 7.6  | 27.6 | 100.0 | -    | 55.5 | 10.2 | C     |
| 157 | Port Moresby | Papua New Guinea | 0.0  | 18.9 | 100.0 | -    | 28.6 | 3.7  | C     |
| 158 | Kampala      | Uganda           | 0.0  | 31.8 | 0.0   | 68.7 | 15.1 | 22.5 | C     |
| 159 | Alexandria   | Egypt            | 0.0  | 45.0 | 0.0   | 11.3 | 7.7  | 30.6 | C     |